Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Teacher's Edition

The Period

MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

Must Decide whether three lengths make a triangle (the two shorter sum to more than the longest) and a right triangle (the converse, on the longest side).

Must not Acute or obtuse by comparing squares (Geometry); sides as expressions; finding a missing side (6.03).

Target I can decide whether three lengths can make a triangle, and whether they make a right triangle.

MTR **MTR.2.1** — Notes I: three lengths as sticks and as one inequality · **MTR.4.1** — boards 3, 5 and 7 split; the sixty-second re-vote · **MTR.6.1** — board 7: a reason checked against the squares, not the sum

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	10 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Add the two shorter sides: MORE than the longest is a triangle; equal lies flat.
2. The third side is between the difference and the sum.
3. Right triangle: the shorter sides' squares add up to the longest side's square.

Slide by Slide

1. **Triangle Inequality Theorem; Converse of the Pythagorean Theorem.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 and Q4 are a pair: $5^2 + 12^2 = 169 = 13^2$, today's converse before it has a name. *1 retrieves 5.10–11 — a unit rate; yesterday was the test, so the band reaches the last lesson (yesterday). 2 retrieves 5.02–03 — area scales by k^2 (last week). 3 retrieves 4.05 — a perfect square's root (Unit 4) (two units back). 4 retrieves MA.7.NSO.1.1 — whole-number powers; the converse adds two squares (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 288 mi 2. 45 cm^2 3. 13 4. 169
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule.** Equal is not enough. Equal lies flat.
6. **Notes.** Two sides, 4 and 9.
7. **Notes.** Circle the longest side first. It is the only one that can be the hypotenuse.
8. **Example 1 — Can the three lengths make a triangle?** Ask which two sides get added before anyone adds: the two shorter, 5 and 8.
9. **Example 1 — Worked.** 5 + 8 = 13, and 13 is less than 14: the two short sides cannot reach across.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 7 + 9 = 16 > 15, so yes — a board that says no saw the big 15 and did not add.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Is it a right triangle?** Ask which side is c before anyone squares: 20, the longest, though it is listed first.
13. **Example 2 — Worked.** The longest side is 20, listed first.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 64 + 121 = 185, not 196, so not right — a board that says 'right' compared 185 with something other than 14^2.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** Can sides of 2 ft, 7 ft and 10 ft make a triangle? → ***No — 2 + 7 = 9 is less than 10.*** *Yes — added 7 + 10 or 2 + 10 [MA.8.GR.1.3: test the two shorter]; 'yes — the ends just do not meet' [Florida B1G-M, GR.1.3: believing the sides need not meet at the corners]*
18. **18–19. Board 2.** Can sides of 6 m, 6 m and 11 m make a triangle? → ***Yes — 6 + 6 = 12 is more than 11.*** *No — 'two sides cannot be equal' [Notes I: only the sum matters]; 'it is not a right triangle, so the rule does not apply' [Florida B1G-M, GR.1.3: the Triangle Inequality holds for every triangle]*
19. **20–21. Board 3.** Which three lengths can make a triangle? → ***B — 5, 11, 8*** *(A) 7 + 7 = 14 counted — equal lies flat [Notes I]; (C) added the first two listed, 3 + 10 [MA.8.GR.1.3: test the two shorter]; (D) added the longest with another, 12 + 4 [MA.8.GR.1.3: test the two shorter] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*
20. **22–23. Board 4.** Two sides of a triangle are 7 cm and 12 cm. Write an inequality for the length x, in centimeters, of the third side. → ***5 < x < 19 — more than 5 cm***

and less than 19 cm. *5 ≤ x ≤ 19 — the flat cases included [Notes II: never equal]; x < 19 — no bottom end [Notes II: long enough to reach]*

24–25. Board 5. Two sides of a triangle are 6 cm and 10 cm. Which inequality shows every possible length x, in centimeters, of the third side? → ***C — 4 < x < 16*** *(A) between the two given sides — the third side can be shorter than both [Notes II: more than the difference]; (B) the flat cases included: at 4 or 16 the sides lie flat [Notes II: never equal]; (D) no bottom end — a third side of 4 or less cannot reach [Notes II: more than the difference] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. Do sides of 8 inches, 15 inches and 17 inches make a right triangle? → ***Yes — $8^2 + 15^2 = 289 = 17^2$.*** *No — compared 8 + 15 = 23 with 17 [Notes III: squares, not sides]*

28–29. Board 7. A triangle has sides of 25 cm, 7 cm and 24 cm. Is it a right triangle? Which answer has the right reason? → ***A — yes, because $7^2 + 24^2 = 25^2$*** *(B) the right verdict for the wrong reason — the Triangle Inequality shows only a triangle [MA.8.GR.1.3: two tests]; (C) 25 taken as a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) added the sides, not their squares [Notes III] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

30–31. Board 8. A triangular sail has sides of 20 ft, 21 ft and 29 ft, and it weighs 12 pounds. Does the sail have a right angle? → ***Yes — $20^2 + 21^2 = 841 = 29^2$.*** *The 12 pounds is not needed. 'Cannot tell without the angles' [Notes III: the sides are enough]*

32–33. Board 9. Ana says sides of 6, 7 and 9 make a right triangle because 6 + 7 > 9. Is she right? Explain. → ***No — 6 + 7 > 9 shows only a triangle; 85 is not 81, so it is not right.*** *'Yes, 13 > 9' — the inequality taken as the converse [MA.8.GR.1.3: two tests]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: a triangle? (no, then yes with two equal sides); 3 votes with the equal case and the wrong pair; 4–5 the third side's range, written then voted. The pattern breaks at 6 — a RIGHT triangle?, squares not sums — and 7 lists the longest side first; 8 has the extra number; 9 is written. The independent set walks the same path.

Misconceptions to Watch

Board 1. Yes — added 7 + 10 or 2 + 10 [MA.8.GR.1.3: test the two shorter]

Board 2. No — 'two sides cannot be equal' [Notes I: only the sum matters]

Board 3. 7 + 7 = 14 counted — equal lies flat [Notes I]

Board 4. 5 ≤ x ≤ 19 — the flat cases included [Notes II: never equal]

Board 5. between the two given sides — the third side can be shorter than both [Notes II: more than the difference]

Board 6. No — compared 8 + 15 = 23 with 17 [Notes III: squares, not sides]

Board 7. the right verdict for the wrong reason — the Triangle Inequality shows only a triangle [MA.8.GR.1.3: two tests]

Board 8. 'Cannot tell without the angles' [Notes III: the sides are enough]

Board 9. 'Yes, 13 > 9' — the inequality taken as the converse [MA.8.GR.1.3: two tests]

Materials Whiteboards and markers. Optional: strips cut to 3, 4, 5, 6 and 9 units for Notes I. Calculators allowed, printed nowhere.

Pythagorean Theorem

Teacher's Edition

The Period

MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

Must Find a missing leg or the hypotenuse of a right triangle from $a^2 + b^2 = c^2$, the hypotenuse found first; an irrational answer exact and, when asked, to the nearest tenth.

Must not Stories (6.04); the coordinate plane (6.05); radicands over 225 or with a square factor; 3-D.

Target I can find a missing side of a right triangle — the hypotenuse, or a leg.

MTR **MTR.6.1** — Notes III and boards 3–4: every leg is checked against c · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.5.1** — Notes I and board 5: one theorem however the triangle is turned

Segment	Slides	Time
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Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	10 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Find the square corner first: the side across from it is c, the longest side.
2. The hypotenuse: square, add, then the root.
3. A leg: subtract the squares — and check it is shorter than the hypotenuse.

Slide by Slide

1. **Pythagorean Theorem.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes III's move before it has a name: $10^2 - 6^2 = 64 = 8^2$, the missing leg of the 6-8-10 triangle Notes II builds. *1 retrieves 6.01–02 — the Triangle Inequality, the last class (yesterday). 2 retrieves 5.10–11 — a unit rate used once (last week). 3 retrieves 4.05 — a perfect square's root (Unit 4) (two units back). 4 retrieves MA.7.NSO.1.1 — whole-number powers; a missing leg subtracts two squares like this (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. No — 4 + 7 = 11 < 12 2. 81 pages 3. 14 4. 64
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the theorem and the picture.** Find the square corner. The side across from it is c.
6. **Notes.** Square, add, then the root. The root is the last step.
7. **Notes.** A leg is shorter than the hypotenuse. Always check.
8. **Example 1 — Find the hypotenuse.** Ask where the square corner is and which side is c before anyone squares.
9. **Example 1 — Worked.** 49 + 81 = 130, not a perfect square: c = √130, between 11 and 12 (121 < 130 < 144), and 11.40 to two places, so 11.4.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 25 + 121 = 146 and √146 ≈ 12.08, so 12.1 — a board with 16 added the legs; a board with 146 stopped before the root.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Find a leg.** Ask which number is c before anyone squares: 15, the longest — so this time a square is subtracted.
13. **Example 2 — Worked.** 15 is c, so 15^2 stands alone: $b^2 + 64 = 225$, $b^2 = 161$, b = √161 ≈ 12.69, so 12.7 — and 12.7 is shorter than 15, as a leg must be.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 121 − 36 = 85 and √85 ≈ 9.22, so 9.2 — a board with 12.5 added the squares (√157), longer than the hypotenuse.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. A right triangle has legs of 9 cm and 12 cm. How long is the hypotenuse? → ***15 cm*** *21 — added the legs, not their squares [MA.8.GR.1.1: the theorem is about squares]*

18–19. Board 2. A right triangle has legs of 4 inches and 9 inches. How long is the hypotenuse, to the nearest tenth? → ***√97 in ≈ 9.8 in*** *13 — added the legs [MA.8.GR.1.1]; 9.9 — rounded twice [Notes II: round once, at the end]*

20–21. Board 3. A right triangle has a hypotenuse of 25 ft and one leg of 20 ft. How long is the other leg? → ***15 ft*** *32.0 — added the squares, longer than the hypotenuse [Notes III: a leg is shorter than c]*

22–23. Board 4. A right triangle has a hypotenuse of 41 m and one leg of 40 m. How long is the other leg? → ***A — 9 m*** *(B) stopped at $b^2 = 81$ and forgot the square root [Math Nation names skipping the final root]; (C) subtracted the sides, 41 − 40 [MA.8.GR.1.1: the theorem is about squares]; (D) added the squares instead of subtracting: about 57.3, longer than the hypotenuse [Notes III: for a leg, subtract] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. The right triangle is drawn turned. Its legs are 5 cm and 7 cm. How long is the hypotenuse, to the nearest tenth? → ***√74 cm ≈ 8.6 cm*** *4.9 — subtracted, taking the 7 as the hypotenuse [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]*

26–27. Board 6. A right triangle's hypotenuse is 20 inches; its legs are 16 inches and x. Which equation finds x? → ***D — and x = 12 in*** *(A) the sides added, not their squares [MA.8.GR.1.1]; (B) the hypotenuse put with the legs [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (C) doubled the 20 instead of squaring it — 20^2 means 20 × 20, not 2 × 20 [Notes I: the theorem is about squares] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A right triangle has a hypotenuse of 14 m and one leg of 9 m. How long is the other leg, to the nearest tenth? → ***√115 m ≈ 10.7 m*** *16.6 — added the squares, longer than the hypotenuse [Notes III]*

30–31. Board 8. A pennant is cut as a right triangle with legs of 8 inches and 11 inches. The fabric costs 6 dollars. How long is its longest side, to the nearest tenth? → ***√185 in ≈ 13.6 in*** *The 6 dollars is not needed. 19 — added the legs [MA.8.GR.1.1]; 'cannot tell' — the longest side is the hypotenuse [Notes I]*

32–33. Board 9. Jo finds a leg of a right triangle. The hypotenuse is 13 cm and the other leg is 5 cm. She writes $\sqrt{(5^2 + 13^2)} \approx 13.9$. What went wrong? How long is the leg? → ***She added the squares; a leg must be shorter than 13 cm. Subtract: the leg is 12 cm.*** *'13.9 is fine' — no check against the hypotenuse [Notes III]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: the hypotenuse, whole then to the tenth; 3–4 a leg, whole then the vote. The pattern breaks at 5 — the triangle turned — and 6 asks only for the equation; 7 is a leg to the tenth; 8 has the extra number; 9 is the check in writing. The independent set: a whole hypotenuse, one to the tenth, a whole leg from a big hypotenuse, a leg to the tenth, a turned triangle, the equation and its value.

Misconceptions to Watch

Board 1. 21 — added the legs, not their squares [MA.8.GR.1.1: the theorem is about squares]

Board 2. 13 — added the legs [MA.8.GR.1.1]

Board 3. 32.0 — added the squares, longer than the hypotenuse [Notes III: a leg is shorter than c]

Board 4. stopped at $b^2 = 81$ and forgot the square root [Math Nation names skipping the final root]

Board 5. 4.9 — subtracted, taking the 7 as the hypotenuse [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]

Board 6. the sides added, not their squares [MA.8.GR.1.1]

Board 7. 16.6 — added the squares, longer than the hypotenuse [Notes III]

Board 8. 19 — added the legs [MA.8.GR.1.1]

Board 9. '13.9 is fine' — no check against the hypotenuse [Notes III]

Materials Whiteboards and markers. $a^2 + b^2 = c^2$ on the board for the period (memorized by the test). Calculators allowed, printed nowhere.

Pythagorean Theorem in Real-World Problems

Teacher's Edition

The Period

MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

Must Find a missing side in a real situation: the right angle located, the hypotenuse named, then the theorem; the answer in the story's unit, rounded once where it measures.

Must not 3-D (a box's space diagonal — the book's 6.4.5); radicands over 225; coordinates (6.05–06).

Target I can find a length in a real situation by finding the right triangle inside it.

MTR **MTR.7.1** — every board is a real object: a ladder, a ramp, a screen, a lawn, a kite · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.6.1** — boards 7 and 9: the answer is checked against the story

Segment	Slides	Time
Settle in; post the learning target	1	1 min
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Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	10 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Sketch the story and mark the square corner before any number.
2. A ladder, a diagonal, a ramp, a kite string: each is the hypotenuse.
3. How high, or how far along the ground, is a leg: subtract.

Slide by Slide

1. **Pythagorean Theorem in Real-World Problems.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is today's first move: every rectangle — a screen, a lawn, a box — has four right angles, so its diagonal is a hypotenuse. *1 retrieves 6.03 — a missing leg: subtract the squares (yesterday). 2 retrieves 6.01–02 — the converse, on the longest side (last week). 3 retrieves 5.04 — a real length from a plan's scale (last unit). 4 retrieves Grade 3–4 (MA.4.GR.1) — a rectangle's corners are right angles; today every diagonal cuts off a right triangle (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 7 ft 2. Yes — 81 + 1,600 = 1,681 = 41^2 3. 14 ft 4. 4
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the four triangles' names.** Sketch it. Mark the square corner. Name c.
6. **Notes — A garden 5 m by 12 m, and a path from corner to corner.** A rectangle's diagonal is always a hypotenuse.
7. **Notes.** How high is a leg. Subtract.
8. **Example 1 — A ramp's length.** Ask where the square corner is before any number: under the high end, where the rise meets the ground.
9. **Example 1 — Worked.** The rise and the run are the legs; the ramp is c.
10. **Your Turn — Same steps, your numbers. Boards up when done.** A screen's corner is square, so the diagonal is c: 49 + 100 = 149 and √149 ≈ 12.21, so 12.2 in — a board with 17 added the sides.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — How high a ladder reaches.** Ask which length is c before anyone squares: the ladder, 12 ft.
13. **Example 2 — Worked.** The ladder is c: h^2 = 144 − 25 = 119, √119 ≈ 10.91, so 10.9 ft — less than 12, as a leg must be.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 169 − 36 = 133 and √133 ≈ 11.53, so 11.5 ft — a board with 14.3 added the squares (√205), higher than the ladder is long.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. A tablet screen is 6 inches wide and 8 inches tall. How long is its diagonal? → ***10 in*** *14 — added the sides [MA.8.GR.1.1: the theorem is about squares]*

18–19. Board 2. A 15-ft ladder leans on a wall. Its foot is 9 ft from the wall. How high up the wall does it reach? → ***12 ft*** *17.5 — added the squares: a 15-ft ladder cannot reach 17.5 ft [Notes III]*

20–21. Board 3. A ramp rises 4 ft over a run of 11 ft. How long is the ramp, to the nearest tenth of a foot? → ***About 11.7 ft*** *15 — added the rise and the run [MA.8.GR.1.1]; '11, the same as the run' — the ramp is the hypotenuse, longer [Notes I]*

22–23. Board 4. A 20-ft ladder reaches 16 ft up a wall. How far from the wall is its foot? → ***A — 12 ft*** *(B) added the squares: about 25.6 ft, farther than the ladder is long*

[Notes III: for a leg, subtract]; (C) subtracted the lengths, 20 − 16 [MA.8.GR.1.1: the theorem is about squares]; (D) stopped at $a^2 = 144$ and forgot the square root [Math Nation names skipping the final root] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

24–25. Board 5. A lawn is 9 m by 12 m. Ben walks along two sides; Ava walks straight across, corner to corner. How much shorter is Ava's walk? → ***6 m shorter*** *15 — Ava's walk, not the difference [the question asks how much shorter]; 21 — Ben's walk*

26–27. Board 6. A kite flies on 30 m of string, straight above a spot 18 m from where the string is held. Which length is the hypotenuse? → ***C — the 30-m string*** *(A) took the ground distance as c — it is a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (B) took the height as c — the height is the unknown leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) did not see the right angle under the kite [Florida B1G-M, GR.1.1: failing to visualize the triangle in a real-world problem] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A box is 7 inches wide and 12 inches long. Will a 14-in umbrella lie flat in it, corner to corner? → ***No — the diagonal is about 13.9 in.*** *'Yes, 19 > 14' — added the sides [MA.8.GR.1.1]; 'Yes, 12 < 14' — compared with one side [Notes II: the diagonal is longest]*

30–31. Board 8. A 10-ft ladder that weighs 25 pounds has its foot 7 ft from a wall. How high does it reach, to the nearest tenth of a foot? → ***About 7.1 ft*** *The 25 pounds is not needed. 12.2 — added the squares (√149): higher than the ladder is long [Notes III]*

32–33. Board 9. A window is 12 ft above the ground. Why can an 11-ft ladder never reach it, wherever its foot stands? → ***The height is a leg — always shorter than the 11-ft ladder.*** *'Move the foot closer' — closer, the height grows toward 11 ft but never reaches it, and the window is at 12 [Notes I: the ladder is c]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: a diagonal and a ladder's height, both whole; 3 a ramp to the tenth; 4 the ladder's foot, voted. The pattern breaks at 5 — two walks and their difference — and 6, which asks only which length is c; 7 is will-it-fit by a tenth; 8 has the extra number; 9 is the reason in writing. The independent set: a diagonal, a ladder's foot, a ramp, along two sides against across, will it fit, a kite's height.

Misconceptions to Watch

Board 1. 14 — added the sides [MA.8.GR.1.1: the theorem is about squares]

Board 2. 17.5 — added the squares: a 15-ft ladder cannot reach 17.5 ft [Notes III]

Board 3. 15 — added the rise and the run [MA.8.GR.1.1]

Board 4. added the squares: about 25.6 ft, farther than the ladder is long [Notes III: for a leg, subtract]

Board 5. 15 — Ava's walk, not the difference [the question asks how much shorter]

Board 6. took the ground distance as c — it is a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]

Board 7. 'Yes, 19 > 14' — added the sides [MA.8.GR.1.1]

Board 8. 12.2 — added the squares (√149): higher than the ladder is long [Notes III]

Board 9. 'Move the foot closer' — closer, the height grows toward 11 ft but never reaches it, and the window is at 12 [Notes I: the ladder is c]

Materials Whiteboards and markers. A meter stick or a ladder picture for Notes I if the room needs it. Calculators allowed, printed nowhere.

Distance on the Coordinate Plane

Teacher's Edition

The Period

MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Must Find the distance between two points: on one line by subtracting (across an axis too), slanted by the right triangle whose legs are read from the coordinates; an irrational answer exact and, when asked, to the tenth.

Must not The distance formula (Clarification 2); radicands over 225; perimeters and maps with several legs (6.06).

Target I can find the distance between two points: subtract on a line, and draw the right triangle when the segment is slanted.

MTR **MTR.2.1** — Notes II–III: one distance as a slanted segment, as a right triangle's legs and as $a^2 + b^2 = c^2$ · **MTR.4.1** — boards 3 and 6 split; the sixty-second re-vote · **MTR.6.1** — Notes I and board 9: a distance across zero checked by counting the squares

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Say these three things out loud today

1. Same y: subtract the x's; same x: subtract the y's — across zero the two pieces add.
2. Slanted: draw the right triangle; its legs come from the coordinates.
3. A distance is never negative, and there is no new formula: the triangle is the method.

Slide by Slide

1. **Distance on the Coordinate Plane.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes I's rule before it is written down: same x, so subtract the y's — and 5 − (−3) is 8, across zero. *1 retrieves 6.04 — a leg in a story (yesterday). 2 retrieves 6.01–02 — the converse (last week). 3 retrieves 5.01 and 5.04 — a real distance from a map's scale (last unit). 4 retrieves MA.6.GR.1.2 — the distance between points with the same x-coordinate (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 15 ft 2. Yes — 144 + 1,225 = 1,369 = 37^2 3. 17 km 4. 8
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Across zero, the two pieces add.
6. **Notes.** Slanted? Draw the triangle. Across, then up.
7. **Notes.** Across: the x's. Up or down: the y's. Then the theorem.
8. **Example 1 — On one line.** Ask what the two points share before anyone subtracts: the x, so the distance is in the y's.
9. **Example 1 — Worked.** Same x, so the y's: 7 − (−2) = 9 — 2 below the axis and 7 above it.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same y: 8 − (−3) = 11 — a board with 5 subtracted 8 − 3.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Slanted.** Ask whether the points share an x or a y: neither — so draw the right triangle first.
13. **Example 2 — Worked.** Across 4 − (−2) = 6, up 9 − 1 = 8; 36 + 64 = 100, so 10 units — whole, because 6, 8, 10 is a triple.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Across 2 − (−5) = 7, up 1 − (−3) = 4; 49 + 16 = 65 and √65 ≈ 8.06, so 8.1 — a board with 11 added the legs.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. How far apart are $(-7, -4)$ and $(-7, 6)$? → ***10 units*** *2 — subtracted 6 − 4 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]*

18–19. Board 2. How far apart are $(-9, 2)$ and $(3, 2)$? → ***12 units*** *6 — subtracted 9 − 3 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]*

20–21. Board 3. What is the distance between (−5, 1) and (3, 1)? → ***B — 8 units***
(A) counted the grid lines, not the spaces [Math Nation names counting gridlines]; (C) subtracted 5 − 3 across zero [Florida B1G-M, GR.1.2: (−2, 0) to (3, 0) is not 1]; (D) −5 − 3: a distance is never negative [Notes I] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

22–23. Board 4. How far apart are $A(-4, -3)$ and $B(5, 9)$? → ***15 units*** *21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]; 'cannot, no triangle' — draw one [Math Nation names 'no triangle drawn']*

24–25. Board 5. How far apart are $C(-2, -5)$ and $D(3, 3)$, to the nearest tenth? → ***√89 ≈ 9.4 units*** *13 — added the legs [MA.8.GR.1.2]; √(1² + 8²) — subtracted 3 − 2 across zero [Florida B1G-M, GR.1.2]*

26–27. Board 6. Which gives the distance between (4, 5) and (−3, −1)? → ***C — √(7² + 6²) = √85*** *(A) 4 − 3 = 1 for the across leg — the sign of −3 dropped [Florida B1G-M, GR.1.2: integer errors across an axis]; (B) one point's coordinates taken as the legs [Math Nation names wrong horizontal and vertical lengths]; (D) added the legs [MA.8.GR.1.2: the theorem, not a sum] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. How far is $(5, -12)$ from the origin? → ***13 units*** *17 — added the legs [MA.8.GR.1.2]; 'the square root of −119' — subtracted 25 − 144 [the theorem adds the legs' squares]*

30–31. Board 8. On a town map in miles, the school is at $(-3, 2)$, the library at $(4, 2)$ and the town hall at $(0, 5)$. How far is the school from the library? → ***7 mi*** *The town hall at (0, 5) is not needed. 1 — subtracted 4 − 3 across zero [Florida B1G-M, GR.1.2]; a distance to the town hall — the wrong pair of points*

32–33. Board 9. Ty says (−3, 0) and (7, 0) are 4 units apart. What did he do? What is the distance? → ***He subtracted 7 − 3. Across zero the pieces add: 10 units.*** *'4 is right' — the axis-crossing error [Florida B1G-M, GR.1.2]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: one line, vertical then horizontal, both across zero; 3 votes on the axis error. The pattern breaks at 4 — slanted, with only the points on the grid — whole, then to the tenth (5), then only the expression (6), then from the origin (7); 8 carries an extra point; 9 is written. The independent set: horizontal, vertical, slanted whole, slanted to the tenth, from the origin, a map.

Misconceptions to Watch

Board 1. 2 — subtracted 6 − 4 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]

Board 2. 6 — subtracted 9 − 3 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]

Board 3. counted the grid lines, not the spaces [Math Nation names counting gridlines]

Board 4. 21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]

Board 5. 13 — added the legs [MA.8.GR.1.2]

Board 6. 4 − 3 = 1 for the across leg — the sign of −3 dropped [Florida B1G-M, GR.1.2: integer errors across an axis]

Board 7. 17 — added the legs [MA.8.GR.1.2]

Board 8. 1 — subtracted 4 − 3 across zero [Florida B1G-M, GR.1.2]

Board 9. '4 is right' — the axis-crossing error [Florida B1G-M, GR.1.2]

Materials Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

Real-World Coordinate-Plane Problems; Perimeter

Teacher's Edition

The Period

MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Must Find the perimeter of a figure on the coordinate plane, each slanted side by its right triangle (a triangle with no right angle split by its height); the straight-line and along-the-grid distances on a map, and how much shorter the straight line is; an irrational total kept exact and rounded once.

Must not The distance formula (Clarification 2); radicands over 225 or with a square factor; combining radicals (√34 + √34 is two lengths added, not 2√34 — Algebra 1).

Target I can find the perimeter of a figure on the coordinate plane, and a straight-line distance on a map.

MTR **MTR.2.1** — Notes I and III: one perimeter as sides on the grid, as right triangles and as a sum · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.7.1** — Notes II and boards 5, 7 and 8: maps, the streets against the straight line

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A perimeter adds every side; a slanted side is found by its own right triangle.
2. On a map, the streets add the legs and the straight line is the hypotenuse.
3. No right angle? The height makes two. Round once, at the end.

Slide by Slide

1. **Real-World Coordinate-Plane Problems; Perimeter.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is today's idea with only straight sides: a perimeter adds every side — 7, 3, 7 and 3, across zero for both. *1 retrieves 6.05 — a slanted distance by its triangle (yesterday). 2 retrieves 6.03 — a hypotenuse from the legs (√53: a whole radicand up to 225, GR.1.1 Clarification 3) (earlier this week). 3 retrieves 5.04 — a real distance from a map's scale (last unit). 4 retrieves MA.6.GR.1.3 — the perimeter of a rectangle on the coordinate plane (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 10 units 2. √53 ≈ 7.3 m 3. 75 mi 4. 20 units
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Every side counts. A slanted side gets its triangle.
6. **Notes.** The streets add the legs. The straight line is the hypotenuse.
7. **Notes.** No square corner? Make two. Round once, at the end.
8. **Example 1 — A perimeter on the grid.** Ask which sides lie on grid lines before anyone measures: DE and EF.
9. **Example 1 — Worked.** DE: 3 − (−3) = 6; EF: 6 − (−2) = 8; DF: 36 + 64 = 100, 10.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Across 4 − (−5) = 9 and up 5 − (−7) = 12, so the slanted side is √225 = 15 and P = 9 + 12 + 15 = 36 — a board with 21 left the slanted side out.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A map: the streets and the straight line.** Ask what 'along the streets' means on this map: on the grid lines — across, then up.
13. **Example 2 — Worked.** Across 4 − (−3) = 7, up 3 − (−2) = 5.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Across −2 − (−5) = 3 and down 4 − (−4) = 8: by the streets 11 blocks, straight √73 ≈ 8.54, so 8.5 — a board with 11 for the straight line added the legs.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **Board 1.** Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$. Find its perimeter. → ***30 units*** *17 — left the slanted side out [Notes I: every side]; 34 — took AC as 12 + 5 [MA.8.GR.1.2: the theorem, not a sum]*
17. **Board 2.** A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$. Find its perimeter, to the nearest tenth. → ***22.8 units*** *13 — left the slanted side out [Notes I]; 22.9 — rounded 9.849 to 9.85 and then to 9.9 [one rounding, at the end — Notes III]; 26 — the slanted side as 4 + 9 [MA.8.GR.1.2]*
18. **Board 3.** Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle. Find its perimeter. → ***42 units*** *40 — took both slanted sides as 13, as if the height split the base in half [the foot is at (1, −6): 5 and 9]; a perimeter with 12 in it — the height is inside the triangle, not a side [B1G-M, GR.1.2: the theorem on a triangle with no right angle]*

22–23. Board 4. What is the perimeter of the triangle with corners (−2, 3), (2, 3) and (2, −4), to the nearest tenth? → ***D — 19.1 units*** *(A) 4 + 7: left the slanted side out [Notes I: every side]; (B) 4 + 7 + 11: the slanted side taken as the sum of the legs [MA.8.GR.1.2: the theorem, not a sum]; (C) 4 + 7 + 65: forgot the square root [B1G-M, GR.1.1: root errors] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. On a map in blocks, the library is at $(-7, -1)$ and the pool at $(5, 4)$. How much shorter is the straight line than the streets? → ***4 blocks shorter*** *13 or 17 — one path, not how much shorter [the question asks a difference]; 30 — added the two paths [the question asks a difference]; 2 across, from 7 − 5 [Florida B1G-M, GR.1.2: integer errors across an axis]*

26–27. Board 6. How long is a diagonal of the rectangle with corners (−3, −4), (6, −4), (6, 8) and (−3, 8)? → ***A — 15 units*** *(B) 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]; (C) 9 + 12 + 9 + 12: the perimeter, not one diagonal [Notes II: the straight line is one segment]; (D) 6 and 8, one corner's coordinates, taken as the legs [Math Nation names wrong horizontal and vertical lengths] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. On a park map in meters, the fountain is at $(-5, 2)$ and the gate at $(4, -3)$. How far is it straight across, to the nearest tenth? → ***√106 ≈ 10.3 m*** *14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]; √26 — 4 − 5 = 1 for the across leg [Florida B1G-M, GR.1.2: integer errors across an axis]*

30–31. Board 8. On a trail map in miles, the camp is at $(-3, -6)$, the lake at $(2, 6)$ and the trailhead at $(-1, 0)$. How far is the camp from the lake, in a straight line? → ***13 mi*** *The trailhead at (−1, 0) is not needed. 17 — added the legs [MA.8.GR.1.2]; a distance to the trailhead — the wrong pair of points*

32–33. Board 9. Mia says the segment from (0, 0) to (3, 1) is 3 units long, because it crosses 3 grid squares. Is she right? Explain, and find its length to the nearest tenth. → ***No: it is the hypotenuse of a triangle with legs 3 and 1, √10 ≈ 3.2 units.*** *'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]; 4 — added the legs [MA.8.GR.1.2]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: a right triangle's perimeter, whole then to the tenth; 3 has no right angle and is split by its height, not in half; 4 votes on the perimeter errors. The pattern breaks at 5 — a map, and how much shorter the straight line is — then a rectangle's diagonal voted (6), a park map to the tenth (7), a map with an extra point (8); 9 is written. The independent set: a perimeter whole, a perimeter to the tenth, a parallelogram, straight against the streets, a map to the tenth, a rectangle's missing corner and its diagonal.

Misconceptions to Watch

Board 1. 17 — left the slanted side out [Notes I: every side]

Board 2. 13 — left the slanted side out [Notes I]

Board 3. 40 — took both slanted sides as 13, as if the height split the base in half [the foot is at (1, −6): 5 and 9]

Board 4. 4 + 7: left the slanted side out [Notes I: every side]

Board 5. 13 or 17 — one path, not how much shorter [the question asks a difference]

Board 6. 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]

Board 7. 14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]

Board 8. 17 — added the legs [MA.8.GR.1.2]

Board 9. 'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]

Materials Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

Similar Triangles; Indirect Measurement

Teacher's Edition

The Period

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

Must Decide similarity from all three side ratios; match the sides of a turned pair; find missing sides with k = copy ÷ original; measure with a shadow, a mirror or a ramp when the words say the triangles are similar.

Must not AA named or proved (Geometry); angle sums (Thread B); a mirror or ramp item that does not say its triangles are similar.

Target I can find a missing side of similar triangles with the scale factor, and use similar triangles to measure what I cannot reach.

MTR **MTR.2.1** — Notes I: one pair as a picture, three ratios and one k · **MTR.4.1** — boards 4 and 5 split; the re-vote · **MTR.7.1** — Notes III, Example 2 and boards 6–8: heights measured without climbing

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Similar triangles have equal angles, and every side is multiplied by the same k.
2. Match by size or by angle, never by place on the page; k = copy ÷ original.
3. Same sun, same angle: shadows, mirrors and ramps make similar triangles.

Slide by Slide

1. **Similar Triangles; Indirect Measurement.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 is today's k: copy ÷ original, 4 ÷ 10 = 0.4 — a copy smaller than the original. *1 retrieves 6.06 — a perimeter, the slanted side by its triangle (√13: a whole radicand up to 225, GR.1.2 Clarification 3) (yesterday). 2 retrieves 6.04 — a ladder: the ladder is c, so subtract (last week). 3 retrieves 5.01 — k = copy ÷ original; a k under 1 shrinks (last unit). 4 retrieves MA.7.GR.1.5 — a scale drawing: every length times one number (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 8.6 units 2. 12 ft 3. k = 0.4 4. 21 ft
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Similar: one k on every side. Match by size, not by place.
6. **Notes.** One matching pair gives k. Every side times k.
7. **Notes.** Same sun, same angle: similar triangles.
8. **Example 1 — A turned pair.** Ask which side of the second triangle matches the 5 before anyone divides: the shortest, 15 — wherever it sits.
9. **Example 1 — Worked.** Sorted: 5, 7, 9 and 15, x, y — the 5 and the 15 are both shortest.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Sorted, the copy is 12, x, 28, so x matches the 5: k = 12 ÷ 3 = 4 and x = 20 — a board with about 46.7 matched by the order in the list (3 with 28).
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A mirror on the ground.** Ask which lengths match: Omar's eye height with the tree's height, and his 4 ft to the mirror with the tree's 18 ft — the words say the two right triangles are similar.
13. **Example 2 — Worked.** Ground to ground: k = 18 ÷ 4 = 4.5, so the tree is 5 × 4.5 = 22.5 ft — a board with about 1.1 used k upside down (4 ÷ 18), one with 19 added 14.
14. **Your Turn — Same steps, your numbers. Boards up when done.** k = 12 ÷ 1.5 = 8, so the flagpole is 4 × 8 = 32 ft — a board with 0.5 used k upside down (1.5 ÷ 12).
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. Are triangles with sides 8, 12, 14 and 12, 18, 21 similar? → ***Yes — every ratio is 1.5*** *'No — the sides are different' [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]; 'yes' from two ratios only [Notes I: all three]*

18–19. Board 2. The two triangles are similar. Find x. → ***x = 24*** *13.5 — matched the 8 with the 18 by place on the page, k = 18 ÷ 8 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 20 — added 12 to the 8 [Notes I: similar means multiplied by one k]*

20–21. Board 3. A triangle has sides of 12, 16 and 20 inches. A similar triangle's longest side is 5 inches. Find its other two sides. → ***3 inches and 4 inches*** *48 and 64 — multiplied by 20 ÷ 5 = 4, k upside down [Notes II: k = copy ÷ original]*

22–23. Board 4. The two triangles are similar (the second is turned). Which side of the second triangle matches the 9? → ***C — the 12*** *(A) the side in the same place on the page — the second triangle is turned [Florida B1G-M, GR.2.4: mismatching corresponding sides]; (B) 16, the longest, for the middle side [Notes I: match by size]; (D) looked for a side equal to the 9 — similar taken as equal [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. Which triangle is similar to one with sides 5, 6 and 8? → ***B — 10, 12, 16*** *(A) added 2 to every side [Notes I: similar means multiplied by one k, not added]; (C) checked two ratios, not three: 15 ÷ 8 is not 2 [Notes I: all three ratios]; (D) kept two sides the same [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow. How tall is the tree? → ***24 ft*** *about 9.4 — height matched to shadow crosswise, 15 × 2.5 ÷ 4 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 16.5 — added 12.5 [similar means multiplied]*

28–29. Board 7. A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar. How high is the ramp 8 m from A? → ***2.4 m*** *1.8 — used the 6 m between the uprights, not the 8 m from A [Math Nation names the overlapping-triangles error]; 6.6 — added 6 [similar means multiplied]*

30–31. Board 8. Lena's eyes are 5 ft up. In a mirror on the ground 4 ft away she sees the top of a 45-year-old flagpole, 16 ft past the mirror. The two right triangles are similar. How tall is the flagpole? → ***20 ft*** *The flagpole's age (45 years) is not needed. 180 or 225 — used the 45 years [the extra number]; 1.25 — divided the other way, 4 × 5 ÷ 16 [Florida B1G-M, GR.2.4: mismatching corresponding sides]*

32–33. Board 9. Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2. Is she right? Explain. → ***No — similar means × one k, not + 2.*** *'Yes — each side grew by 2' [Notes I: similar means multiplied by one k]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–3: similar or not by three ratios, a turned pair's missing side (k = 3), then a k under 1; 4 votes on matching in a turned pair, 5 on which triangle is similar. The pattern breaks at 6 — a shadow — then a ramp's two triangles from their shared corner (7) and a mirror with an extra number (8); 9 is written. The independent set: similar or not (it is not), a missing side matched by the middle sides, a k under 1, a shadow, a nested pair from the shared corner (with its picture), which is similar.

Misconceptions to Watch

Board 1. 'No — the sides are different' [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]

Board 2. 13.5 — matched the 8 with the 18 by place on the page, k = 18 ÷ 8 [Florida B1G-M, GR.2.4: mismatching corresponding sides]

Board 3. 48 and 64 — multiplied by 20 ÷ 5 = 4, k upside down [Notes II: k = copy ÷ original]

Board 4. the side in the same place on the page — the second triangle is turned [Florida B1G-M, GR.2.4: mismatching corresponding sides]

Board 5. added 2 to every side [Notes I: similar means multiplied by one k, not added]

Board 6. about 9.4 — height matched to shadow crosswise, 15 × 2.5 ÷ 4 [Florida B1G-M, GR.2.4: mismatching corresponding sides]

Board 7. 1.8 — used the 6 m between the uprights, not the 8 m from A [Math Nation names the overlapping-triangles error]

Board 8. 180 or 225 — used the 45 years [the extra number]

Board 9. 'Yes — each side grew by 2' [Notes I: similar means multiplied by one k]

Materials Whiteboards and markers. Calculators allowed, printed nowhere. The Unit Review, one per student, goes home today.

Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

Teacher's Edition

The Period

MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.

Must Name adjacent, vertical, complementary and supplementary pairs (a linear pair is supplementary); find a missing angle from each; write and solve the equation when the angles are expressions, and report the angle the question asks for.

Must not Parallel lines and transversals (Geometry); angle sums of polygons (MA.8.GR.1.6); reflex angles.

Target I can name an angle pair and use it to find a missing angle, with an equation when the angles are expressions.

MTR **MTR.2.1** — Notes II–III: one pair as a picture, a name and an equation · **MTR.4.1** — boards 4 and 7 split; the re-vote · **MTR.5.1** — Notes III and boards 5–6: the same equation shape behind every pair

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	12 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Adjacent angles share a side; vertical angles sit across a crossing and are equal.
2. Complementary makes 90 — a corner; supplementary makes 180 — a straight line.
3. Name the pair, write the equation, solve — then put x back: the question asks for the angle.

Slide by Slide

1. **Angle Pairs: Supplementary, Complementary, Vertical, Adjacent.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is today's 'supplementary' before it has a name: a straight angle is 180°, so the other is 50°. *1 retrieves 6.07 — a shadow; yesterday was the test, so the band reaches the last lesson (yesterday). 2 retrieves 6.05 — same y, across zero (last week). 3 retrieves 5.10–11 — a unit rate (last unit). 4 retrieves MA.4.GR.1.3 — angle measures add; a straight angle is 180° (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 27 ft 2. 9 units 3. 42 4. 50°
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the names and the picture.** Next to, sharing a side: adjacent. Across: vertical, and equal.
6. **Notes.** Complementary: corner, 90. Supplementary: straight, 180.
7. **Notes.** Name it, write it, solve it, put x back.
8. **Example 1 — A right angle, split.** Ask what the square mark says before anyone writes: the two angles make 90 — complementary.
9. **Example 1 — Worked.** Complementary: 3x − 1 + 2x + 6 = 90, so 5x + 5 = 90, x = 17.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 4x + x + 10 = 90, 5x = 80 and x = 16, so 64° and 26° — a board with 16 stopped at x; one with 34 used 180.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Vertical angles.** Ask what vertical angles do: they are equal — so the two expressions are set equal.
13. **Example 2 — Worked.** Vertical, so equal: 5x − 12 = 2x + 18; take 2x off both sides, 3x − 12 = 18, so 3x = 30 and x = 10.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 4x + 3 = x + 39, and taking x off both sides gives 3x + 3 = 39, so x = 12 and each angle is 51° — a board with 12 reported x.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** What is the complement of 34°? → ***56°*** *146° — used 180 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]*
17. **18–19. Board 2.** What is the supplement of 112°? → ***68°*** *−22 or 'none' — used 90 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]*
18. **20–21. Board 3.** Two lines cross. One angle is 47°. Find a and b. → ***a = 47°, b = 133°*** *a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum]*
19. **22–23. Board 4.** $\angle P$ and $\angle Q$ are complementary, and $m\angle P = 28^\circ$. What is $m\angle Q$? → ***B — 62°*** *(A) 180 − 28: supplementary for complementary [Florida B1G-M, GR.1.4: complementary and supplementary swapped]; (C) took the two as equal — that is vertical angles [Notes I]; (D) 90 + 28: added instead of subtracting [Notes II: the two add*

to 90] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

24–25. Board 5. A linear pair measures $(3x+15)^\circ$ and 105°. Find x. → ***x = 20*** *30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]; −10 — used 90 [Florida B1G-M, GR.1.4]*

26–27. Board 6. Vertical angles measure $(6x-20)^\circ$ and $(2x+28)^\circ$. Find the measure of each angle. → ***52°*** *12 — reported x as the angle [Math Nation names reporting x]; 21.5 — added them to 180 [vertical angles are equal, not supplementary]*

28–29. Board 7. In the figure, lines AD and BE cross at O, and ray OC is drawn. Which two angles are adjacent? → ***C — angles AOB and BOC*** *(A) a vertical pair: across the crossing, no shared side [Notes I]; (B) angle AOB lies inside angle AOC — the two overlap [Notes I: no overlap]; (D) these share only the vertex O, not a side [Notes I: a shared side] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

30–31. Board 8. Above a straight line a ray makes angles of 38° and x°; below it another ray makes an angle of 50°. Find x. → ***x = 142*** *The 50° angle below the line is not needed. 92 — used the 50° (180 − 38 − 50) [the extra number]; 52 — used 90 [Florida B1G-M, GR.1.4]*

32–33. Board 9. Can two vertical angles be complementary? Always, sometimes or never? Explain. → ***Sometimes — only when each is 45°.*** *'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]; 'always' [Florida B1G-M, GR.1.4]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: a complement, then a supplement; 3 reads a crossing (vertical and a linear pair at once); 4 votes on 90 against 180. The pattern breaks at 5 — a linear pair with x — then vertical angles with x where the angle is asked (6), which pair is adjacent voted (7), and a picture with one angle too many (8); 9 is written. The independent set: a complement, a supplement, a crossing, a linear pair with x, vertical angles with x, a pair named and used.

Misconceptions to Watch

Board 1. 146° — used 180 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]

Board 2. −22 or 'none' — used 90 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]

Board 3. a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum]

Board 4. 180 − 28: supplementary for complementary [Florida B1G-M, GR.1.4: complementary and supplementary swapped]

Board 5. 30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]

Board 6. 12 — reported x as the angle [Math Nation names reporting x]

Board 7. a vertical pair: across the crossing, no shared side [Notes I]

Board 8. 92 — used the 50° (180 − 38 − 50) [the extra number]

Board 9. 'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]

Materials Whiteboards and markers. Calculators allowed, printed nowhere.

Interior and Exterior Angles of a Triangle

Teacher's Edition

The Period

MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.

Must Find a missing interior angle with the Triangle Sum Theorem; find an exterior angle, or a remote interior angle, with the Exterior Angle Theorem; write and solve the equation when the angles are expressions.

Must not Angle sums of polygons (MA.8.GR.1.6); parallel lines and transversals; reflex angles (the angle around the outside is a named error, not an answer).

Target I can find a missing angle of a triangle, inside or outside, with the Triangle Sum and the Exterior Angle Theorem.

MTR **MTR.2.1** — Notes I–II: 180° as torn-off corners, as a sum and as a linear pair · **MTR.4.1** — boards 4 and 7 split; the re-vote · **MTR.5.1** — Notes II: the theorem from the triangle sum and the linear pair together

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	12 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The three interior angles of a triangle add to 180°.
2. An exterior angle equals the two remote interior angles added — not the angle beside it.
3. With expressions: write the sum or the theorem, solve, put x back, check 180.

Slide by Slide

1. **Interior and Exterior Angles of a Triangle.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes I before it is said: three angles along a straight line make 180°. *1 retrieves T-B1 — supplementary: the two make 180 (yesterday). 2 retrieves 6.05 — a slanted distance by its triangle (last week). 3 retrieves 5.10–11 — a unit rate (last unit). 4 retrieves MA.4.GR.1.3 — angle measures add; three angles along a straight line make 180° (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 97° 2. 10 units 3. $0.75 4. 70
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Inside a triangle, the three make 180.
6. **Notes.** Outside equals the two far ones added.
7. **Notes.** Write the sum, solve, put x back, check 180.
8. **Example 1 — The third angle.** Ask what the three angles of any triangle add to before anyone subtracts: 180.
9. **Example 1 — Worked.** 58 + 71 = 129, and 180 − 129 = 51° — a board with 231 used 360; one with 129 stopped at the sum.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 39 + 84 = 123, so 180 − 123 = 57° — a board with 123 stopped at the sum.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — An exterior angle.** Ask which two angles are remote — not touching the exterior angle: the 64° and the 49°.
13. **Example 2 — Worked.** The Exterior Angle Theorem: x = 64 + 49 = 113°.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 72 + 35 = 107° — a board with 73 found the interior angle beside it.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. Two angles of a triangle are 47° and 68°. Find the third angle. → ***65°*** *115 — stopped at the sum [Math Nation names misremembering the sum]; 245 — used 360 [the Triangle Sum is 180]*

18–19. Board 2. An isosceles triangle's two base angles are equal. Each is 52°. Find the third angle. → ***76°*** *128 — used one 52 [the two base angles are equal: both count]; 52 — took all three as equal [isosceles: two equal, not three]*

20–21. Board 3. A triangle's remote interior angles are 38° and 77°. Find the exterior angle x. → ***115°*** *65 — the interior angle beside it, 180 − 115 [Notes II: the exterior angle is not the interior angle beside it]; 295 — the angle around the outside, 360 − 65 [Florida B1G-M, GR.1.5: the reflex angle]*

22–23. Board 4. A triangle's angles are 45°, 60° and 75°. What is the exterior angle beside the 75°? → ***B — 105°*** *(A) equals the adjacent interior angle [Notes II: the exterior angle is not the interior angle beside it]; (C) the angle around the outside, 360 − 75 [Florida B1G-M, GR.1.5: the exterior angle taken as the reflex angle]; (D) only one remote angle [the Exterior Angle Theorem adds both] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. An exterior angle of a triangle is 130°. One remote interior angle is 55°. Find the other remote interior angle. → ***75°*** *185 — added the exterior and the remote angle [the theorem: the remote angles add TO the exterior one]; 50 — the interior angle beside the exterior one [Notes II: the exterior angle is not the interior angle beside it]*

26–27. Board 6. A triangle's angles are x°, $3x^\circ$ and $(2x+12)^\circ$. Find each angle. → ***28°, 84° and 68°*** *28 alone — reported x, one angle of three [Math Nation names reporting x]*

28–29. Board 7. An exterior angle measures (5x + 4)°; its remote interior angles measure 52° and (3x)°. Which equation finds x? → ***C — 5x + 4 = 52 + 3x*** *(A) the exterior angle and one remote angle taken as a linear pair [Notes II: the exterior angle's linear pair is the interior angle beside it]; (B) one remote angle only [the theorem adds both]; (D) the exterior angle used as an interior angle in the Triangle Sum [Math Nation names writing the equation for the wrong relationship] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

30–31. Board 8. A triangle has angles of 41° and 66°, and its longest side is 9 cm. Find its third angle. → ***73°*** *The 9-cm side is not needed. 64 — subtracted the 9 as well [the extra number]; 107 — stopped at the sum*

32–33. Board 9. Why is an exterior angle of a triangle always bigger than either of its remote interior angles? Explain. → ***It is the two remote angles added, each more than 0°.*** *'Because it is on the outside' [no reason]; 'it is not always bigger' [the remote angles are each more than 0°]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: the third angle, then an isosceles triangle; 3 is an exterior angle; 4 votes on the angle beside the 75°. The pattern breaks at 5 — a remote angle found backwards from the exterior one — then three expressions (6), which equation fits the words voted (7), a side length that is not an angle (8); 9 is written. The independent set: a third angle, an isosceles triangle's base angles, an exterior angle, a remote angle, three expressions, an equation from the theorem.

Misconceptions to Watch

Board 1. 115 — stopped at the sum [Math Nation names misremembering the sum]

Board 2. 128 — used one 52 [the two base angles are equal: both count]

Board 3. 65 — the interior angle beside it, 180 − 115 [Notes II: the exterior angle is not the interior angle beside it]

Board 4. equals the adjacent interior angle [Notes II: the exterior angle is not the interior angle beside it]

Board 5. 185 — added the exterior and the remote angle [the theorem: the remote angles add TO the exterior one]

Board 6. 28 alone — reported x, one angle of three [Math Nation names reporting x]

Board 7. the exterior angle and one remote angle taken as a linear pair [Notes II: the exterior angle's linear pair is the interior angle beside it]

Board 8. 64 — subtracted the 9 as well [the extra number]

Board 9. 'Because it is on the outside' [no reason]

Materials Whiteboards and markers; one paper triangle to tear for Notes I, if there is time. Calculators allowed, printed nowhere.